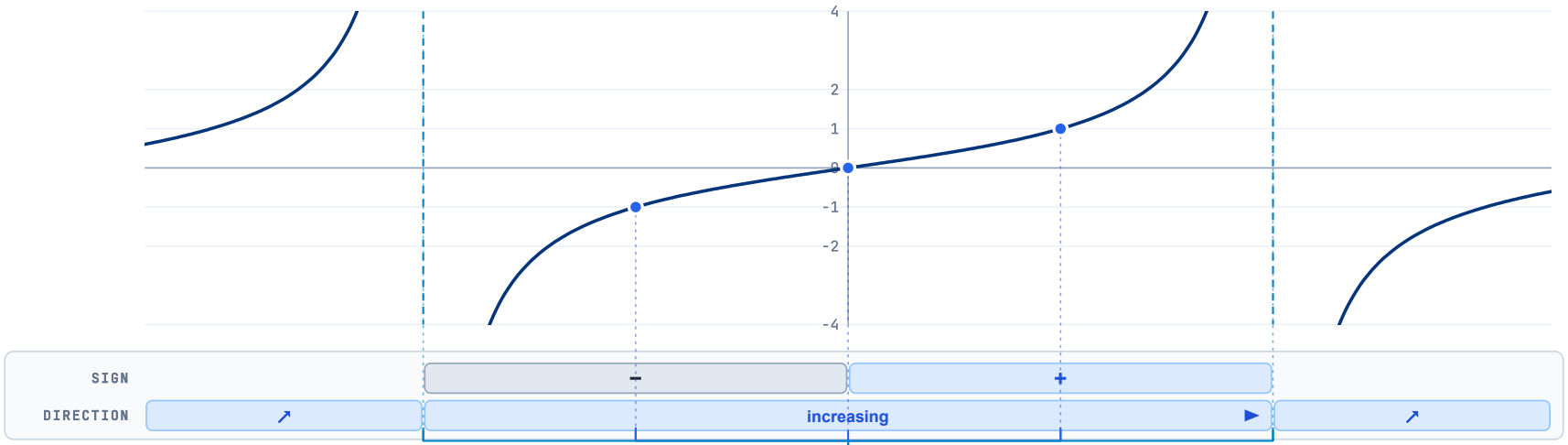
PRINCIPAL PERIOD

Profile of the tangent $y = \tan(x)$

Π PERIOD · ODD SYMMETRY



KEY POINTS

1 ZERO
$(0, 0)$

2 QUARTER POINTS
$(\pm\pi/4, \pm 1)$

3 ASYMPTOTES
$x = \pm\pi/2$

GLOBAL

PERIOD
π, not 2π

RANGE
$(-\infty, \infty)$
unbounded, so no amplitude

ALL ASYMPTOTES
$x = \frac{\pi}{2} + n\pi$
where $\cos x = 0$

SYMMETRY
odd

EACH BRANCH
strictly increasing
concavity flips at each zero